Temporal Emotional Drift Analysis in Digital Conversations for Early Mental Health Risk Detection

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***Abstract*—In recent years, the prevalence of mental disorders has become increasingly widespread; however, identifying the initial stages of such conditions is quite challenging because of the existence of various obstacles such as social stigma, lack of psychological assistance and the use of subjective methods. Since the Internet usage is increasing significantly, people tend to share their feelings by means of social media posts, instant messages and discussions in specialized forums. All these digital communication tools provide useful information concerning emotional changes which could be used to track users' state of mind. This paper presents the concept of the Temporal Emotional Drift Detection which is developed to research gradual and abrupt shifts in the emotional coloring of prolonged dialogues. As opposed to existing approaches, where each message was considered independently, the new method suggests examining how people experience and express their emotions during the course of the conversation to reveal any emerging problems with mental health.**

***Index Terms*—Mental health monitoring; Temporal emotional drift; Digital conversations; Social media analysis; Early risk detection; Natural language processing; Emotion dynamics**

I. Introduction

Mental health conditions such as depression, anxiety, and suicidal ideation are recognized as major contributors to the global burden of disease, affecting millions of individuals across different age groups, professions, and socio-economic backgrounds. These conditions significantly influence emotional well-being, interpersonal relationships, productivity, academic performance, and overall quality of life. In severe cases, untreated mental health disorders may lead to self-harm, substance abuse, or suicide. Although awareness regarding mental health has improved in recent years, early diagnosis and timely intervention remain challenging due to social stigma, lack of mental health professionals, limited accessibility to clinical care, and dependence on self-reporting mechanisms. As a result, many individuals experiencing psychological distress remain undiagnosed until symptoms become severe. Therefore, there is an increasing need for scalable and non-intrusive methods capable of continuously monitoring emotional and behavioral patterns to support early mental health risk detection.

The rapid growth of digital communication technologies has transformed the way individuals express emotions and interact socially. Social media platforms, online forums, blogs, and instant messaging applications have become integral parts of everyday communication. Users frequently share personal thoughts, emotions, opinions, and life experiences through text-based interactions, often revealing emotional states either explicitly or implicitly. Unlike conventional clinical assessments that occur periodically in controlled environments, digital conversations are generated continuously in naturalistic settings and provide real-time insights into evolving psychological conditions. These online interactions create large-scale longitudinal textual data containing emotional trajectories that can be analyzed computationally to understand behavioral and affective changes over time [3], [4].

Recent developments in Natural Language Processing (NLP), machine learning, and deep learning have enabled automated analysis of textual data for mental health assessment [1], [2], [5]. Several studies have applied transformer-based architectures, recurrent neural networks, hierarchical attention models, and contextual language embeddings to detect depression, anxiety, stress, and suicidal tendencies from social media posts and user-generated content [1], [2], [5], [6]. These approaches have demonstrated promising performance in identifying emotional distress and psychological risk factors from textual information. In addition, sentiment analysis and emotion recognition techniques have been widely used to classify positive, negative, or neutral emotional states within online conversations [4], [8].

However, many existing approaches treat social media posts or messages as isolated instances and primarily focus on static classification tasks [1], [5]. Such methods often fail to capture the temporal and evolving nature of human emotions. Emotional conditions generally develop progressively through subtle behavioral and linguistic changes rather than appearing abruptly. A user may gradually transition from neutral or mildly negative emotions toward persistent distress over weeks or months. Analyzing individual posts independently may therefore overlook critical early warning signs associated with mental health deterioration. Furthermore, temporary emotional fluctuations caused by daily events can be misinterpreted as severe psychological conditions when temporal context is ignored.

To address these limitations, this research proposes a framework called Temporal Emotional Drift Analysis for modeling emotional evolution within time-ordered digital conversations. The proposed framework focuses on identifying gradual emotional drift and abrupt transitions in emotional polarity and intensity across longitudinal textual interactions. By incorporating temporal context windows, sequential representations, and drift-aware indicators, the approach aims to distinguish short-term emotional fluctuations from sustained negative behavioral trends. The framework seeks to provide a scalable, privacy-aware, and non-intrusive mechanism for continuous mental health monitoring and early risk identification. Such temporally aware emotional analysis can support psychologists, healthcare professionals, and digital well-being systems in enabling timely intervention and preventive mental health care.

II. Literature Review

Recently, there is a strong interest in applying Natural Language Processing (NLP) and deep learning techniques to detect mental health using social media data. Some studies have aimed to identify depression, anxiety, suicidal ideation and emotional distress from textual conversations shared on digital platforms [1], [2], [5]. Zhou and Mohd proposed a framework of deep learning based on BERT and BiLSTM for the detection of depression from social media text [1]. Their approach included emoji normalization, slang replacement and emotion scoring to enhance contextual understanding of informal online language. The model was able to achieve 88.95% accuracy on the data set of 20,000 tweets. The framework performed well for classification but was primarily designed for static text analysis and did not take into account the emotional evolution over time or behavioural changes at the user level.

Likewise, Zogan et al. proposed a Hierarchical Convolutional Attention Network for depression detection from Twitter data [2]. Their model took into account tweet history at user level and achieved 90% accuracy. The study highlighted the importance of contextual relationships between posts. However, it provided little analysis of emotional drift and temporal evolution in conversations.

Some studies on temporal emotion analysis during the COVID-19 pandemic have also been made. Yu et al. conducted sentiment time-series analysis on more than 800,000 Weibo posts to analyze public emotions [3]. While their work successfully captured the stage-wise emotional changes during the pandemic, the lack of predictive modeling hindered the early identification of mental health risks. Similarly, Adikari et al. used NLP techniques, word embeddings and Markov models to analyze self-reported emotional information from social media conversations [4].

However, while these techniques had outstanding results in predicting transitions between different states of emotions, none of them stressed longitudinal tracking of emotions from users. Applying transformers for predicting depression led to achieving high levels of efficiency. Specifically, Zhang et al. developed a hierarchical transformer-based technique that was aimed at predicting depression with the use of a Sina Weibo data set, which was performed with 95.46% accuracy [5]. Although the approach revealed excellent skills in terms of language and context analysis, the primary benefit of the approach was static classification of emotions instead of longitudinal emotional monitoring. In another paper, Kim et al. applied BERT embeddings and clustering in order to analyze millions of Reddit posts and identify various mental diseases [6]. Despite the contribution of the paper to the research in mental disorders, it remained cross-sectional.

The use of techniques for analyzing emotions has also been researched for the purpose of suicide prediction. Suicidal behavior has been analyzed by Garc´ıa-Mart´ınez et al. via emotional valence correlation models based on Spanish tweets [7]. While there was some correlation between emotional expression and suicide risk observed by them, there was a drawback due to dependence on manual tagging, making scale-up difficult. Choudrie et al. have used RoBERTa based emotion classification models in text analysis of COVID-19 data and achieved an accuracy of 80.33% [8], yet focusing more on trends than emotional evolution.

Hybrid DL techniques have been shown to be promising as well. Mansoor and Ansari proposed a DL hybrid model combining CNN and LSTM models for mental health effect assessment using Social Media big data and achieved an accuracy of 99.42% [9]. This approach was limited to the context of COVID-19, even though it displayed impressive predictive capabilities.

In combination, all past studies demonstrate that NLP and DL methods can be used effectively to detect mental health conditions. It is worth mentioning, nevertheless, that most existing methods rely primarily on sentiment classification and post-level analysis. On the other hand, there has not yet been much research conducted concerning the modelling of emotions during a time-ordered conversation. As such, in order to fill the observed gap, the Temporal Emotional Drift Analysis concept has been introduced.

III. Scope of work

This project focuses on designing and evaluating a Temporal Emotional Drift Analysis framework for early mental health risk detection using digital conversations and social media interactions. The proposed system analyzes emotional patterns expressed in time-ordered textual data to identify gradual emotional deterioration associated with depression, anxiety, and psychological distress. Unlike conventional sentiment analysis approaches that treat posts independently, the framework emphasizes temporal emotional evolution by tracking changes in emotional polarity and intensity over time.

The proposed methodology integrates Natural Language Processing techniques, contextual language embeddings, sentiment analysis, and deep learning models for sequence-based emotional analysis [1], [2], [5]. BERT embeddings are used to capture semantic and contextual information from textual conversations [10], while a BiLSTM-based sequence modeling approach is employed to learn temporal dependencies and emotional progression patterns [11]. Temporal emotional drift is incorporated as an additional feature to detect both gradual and abrupt emotional transitions within longitudinal user interactions.

The framework is evaluated using labeled social media datasets containing user posts and timestamps. The system performs preprocessing operations such as text cleaning, emoji normalization, slang replacement, and tokenization to improve robustness against noisy social media data [1], [14]. Performance evaluation is carried out using metrics including accuracy, precision, recall, F1-score, and AUC to assess the effectiveness of early mental health risk identification.

The scope of this work is limited to text-based digital conversations and social media data analysis. The proposed framework does not include clinical diagnosis or real-time deployment in healthcare systems; however, the methodology can be extended to multilingual datasets, multimodal behavioral analysis, and advanced transformer-based temporal architectures for broader mental health monitoring applications.

IV. Data Description

The dataset used in this study was obtained from Kaggle and contains approximately 5,000 labeled social media posts for depression detection. The dataset includes attributes such as post text, user id, post created, and label. The dataset structure follows the depression detection framework reported by Zhou and Mohd [1].

To enable longitudinal behavioral analysis, posts belonging to the same user are grouped using user id and chronologically arranged based on the post created timestamps. Each user's posts are then represented as a time-ordered sequence. This temporal restructuring allows the proposed system to capture gradual emotional transitions and evolving behavioral patterns instead of analyzing individual posts independently.

V. Methodology

A. Dataset Description and Temporal Structuring

We derived the dataset from Kaggle which contains 5,000 posts labeled for depression detection. Posts contain information such as text of post, user id when the post was created, and a label. The data set was constructed using a framework for depression detection proposed by Zhou and Mohd [1]. This framework is different than looking at each post individually and using only text. It takes into account how the users information changes through the years. We group the posts by user, placed them in order by the time they were posted, and then looked at the posts as a series of posts that has occurred over time, so we can observe how the users' emotions and behavior change over time.

B. Data Preprocessing

People's language in the media is random, since they could use numerous slang words, emoticons, and abbreviations. Moreover, they tend to violate the grammar rules while posting on their accounts. Therefore, in order to use this information and train a model based on this data, it was necessary to perform some preprocessing steps. Firstly, we removed those elements from text posts which were redundant and did not add value to the information, such as hyperlinks, hashtags, and punctuation marks. Secondly, we converted emoticons into words which would reflect the meaning of symbols used by people to preserve emotions conveyed in posts. Thirdly, slang words and abbreviations were transformed into regular words following the slang dictionary. Lastly, we tokenized our data with the help of the BERT Tokenizer [10].

C. Feature Representation

Each media post was converted into a feature representation that contains several different dimensions. First, we made use of the tool called BERT for making embeddings that represent the context and semantics behind every word [10]. Such embeddings are essentially multi-dimensional images depicting the relations between each of the words. Furthermore, the sentiment analysis of every post was performed with the help of the TextBlob library, which returned a score from the interval [-1,1] indicating whether there were any emotions present and what type of emotions they were [15]. More specifically, the negative score indicates that the post had emotions, while the positive score indicates positive emotions. In some cases, the features like the rate of retweets and number of followers were also considered.

D. Temporal Emotional Drift Modeling

The proposed model incorporates the notion of temporal emotional drift to account for the temporal change of emotional states.

$$\Delta E_t = E_t - E_{t-1} \tag{1}$$

$\Delta E > 0$ reflects improvement in emotional states, whereas $\Delta E < 0$ reflects emotional degradation. The initial drift value will be set to:

$$\Delta E_1 = 0 \tag{2}$$

The proposed model allows for the detection of gradual emotional degradation and abrupt emotional transitions as opposed to using traditional classification methods.

E. Sequence Modeling

The proposed architecture processes user-level temporal sequences using a Bidirectional Long Short-Term Memory (BiLSTM) neural network [11]. The input feature vector in the sequence will be specified as follows at each time step t:

$$X_t = [B_t,\ E_t,\ \Delta E_t,\ BF_t] \tag{3}$$

where B denotes the BERT embeddings, E represents the emotion score, ΔE denotes the temporal emotional drift, and BF represents the optional behavioral features.

The BiLSTM architecture was selected because it can learn contextual dependencies and bidirectional temporal relationships within emotional sequences. By processing information from both past and future contexts, the model can effectively identify trends in emotional development throughout discussions. Additionally, an attention-based mechanism can be incorporated to capture significant emotional shifts and enhance the model's interpretability.

F. Model Training

In this work, the training of our model is going to happen with the help of the Adam optimizer. As for the value of learning rate, it will be equal to 1 times 10^(-5). At that, the training will happen by batches. We are going to use a batch size equal to 32. Number of epochs of model training will be in the range from 3 to 5. The loss function which will be used is going to be the Binary Cross-Entropy function. With the help of some measures, we increase the ability of our model to generalize on data. For instance, we may apply dropout and batch normalization.

G. Evaluation Metrics

When it comes to evaluation of the model, it will happen with the help of such metrics as Accuracy, Precision, Recall, F1-score and AUC. It will be important to analyze not only the model classification ability but its ability to identify the early signs of depression.

H. Proposed Architecture

Our model differs from models with transformers used for assessing mental conditions. Our architecture is based on analyzing dynamics and sentiment. Thus, our proposed architecture can analyze the discussion and emotions of users. Our model represents a pipeline with several tasks. First, it gathers media posts of users and organizes them in chronological order. The next task is cleaning texts by replacing emojis and slang with common language. After that, BERT is applied for the analysis of meaning [10]. Afterwards, the system assesses the sentiments of the text with sentiment analysis and calculates dynamics of emotions. In the end, using all collected data, our model learns patterns and applies them to predict the depression risk.

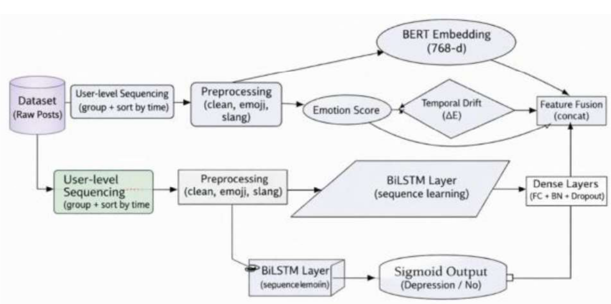

Fig. 1. Proposed Architecture for Temporal Emotional Drift Analysis

I. Architecture Description

We use a kind of network called BiLSTM to learn the patterns [11]. We use something called dense layers to make the predictions. The dense layers have a sigmoid activation function which helps us get a prediction that's, between 0 and 1. This is what we want for depression risk prediction.

VI. Result Analysis

A. Experimental Results

In order to validate the framework, we conducted experiments using a Kaggle social media dataset of mental health that includes approximately 5,000 labeled posts. In this experiment, we utilized a typical 80:20 splitting approach where the bigger part of the dataset was used for training while the other fifth was left aside for testing purposes. This particular system combined three major elements: the BERT embeddings for interpreting text content, temporal emotional drifting for detecting the changes of emotions, and BiLSTM sequences to detect emotions patterns, with the overall aim being depression prediction.

TABLE I
Classification Performance Metrics

Metric	Non-Depression	Depression
Precision	0.80	0.81
Recall	0.80	0.82
F1-Score	0.80	0.82

From the results, it can be observed that both the categories have decent predictive accuracy with a slight edge shown by the recall of the depression category. This indicates that the system has managed to identify the depressive behavior without having committed any false negatives.

B. Loss Curve Analysis

During the process of training, it was observed that the loss value kept on decreasing, indicating successful convergence of BERT + BiLSTM methodology.

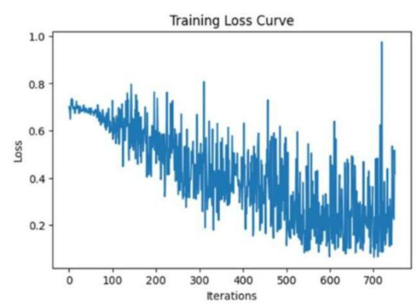

Fig. 2. Training loss curve of the proposed Temporal Emotional Drift Analysis model

C. Dataset Distribution Analysis

The dataset distribution analysis revealed that the number of depressive and non-depressive samples was nearly balanced. Balanced data distribution helps reduce classification bias and improves the generalization capability of the proposed framework.

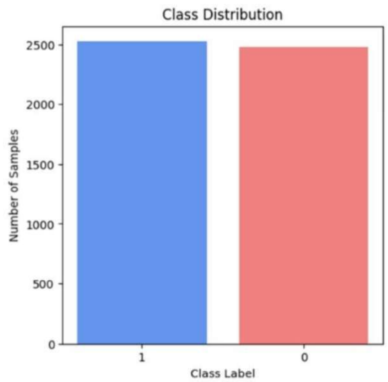

Fig. 3. Distribution of depression and non-depression samples in the dataset

D. Confusion Matrix Analysis

The confusion matrix demonstrates the classification capability of the proposed framework. Most samples were correctly classified into their respective categories, while only a limited number of misclassifications were observed. The results indicate that the framework effectively distinguishes depressive emotional patterns from normal conversational behavior.

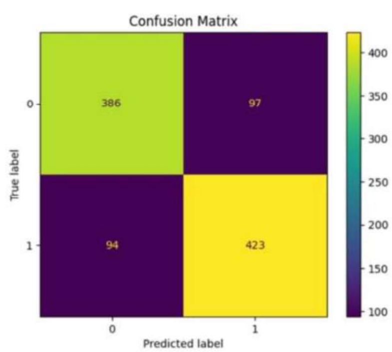

Fig. 4. Confusion matrix of the proposed depression detection model

E. Precision, Recall, and F1-Score Analysis

The evaluation metrics further validate the robustness of the proposed framework. Precision measures the correctness of depressive predictions, recall evaluates the ability to identify actual depressive samples, and F1-score provides the harmonic balance between precision and recall. The obtained metric values indicate that the proposed model performs consistently across all major evaluation measures.

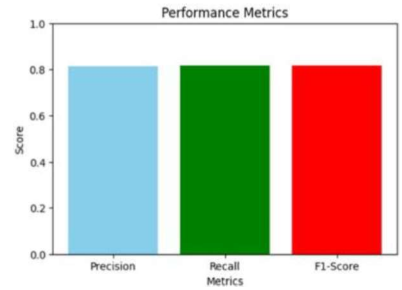

Fig. 5. Comparison of precision, recall, and F1-score metrics

F. Discussion

The results show that incorporating temporal emotional drift analysis improves mental health risk detection from digital conversations. The combination of BERT embeddings and BiLSTM sequence modeling effectively captures both semantic and temporal emotional patterns. Although the achieved accuracy is slightly lower than some large-scale models, the proposed framework provides a meaningful temporal perspective for early mental health risk identification. Future improvements may include larger datasets, attention mechanisms, and transformer-based temporal models.

VII. Conclusion

In this paper, a Temporal Emotional Drift Analysis method is presented that focuses on detecting signs of potential mental health risks by studying conversations and activities performed on social media platforms. The approach combines BERT contextual embeddings, sentiment scoring, modeling of emotional drift over time, and BiLSTM sequence learning for assessing changes in emotions over time periods.

In contrast to static sentiment analysis, the approach allows analyzing slow changes in emotions, together with behavioral patterns, within the history of user conversations. The experiments reveal that the incorporation of temporal emotional drift enhances the performance of the system when it comes to classifying text-based communication in terms of depression presence.

Combining contextual embedding of language and temporal modeling of sequences makes it possible to achieve strong results with respect to both semantic and emotional relationships. Further improvements in achieved accuracy can be made by employing larger datasets and neural network transformers. However, the suggested technique shows promise for creating an approach to mental health assessment that does not employ invasive means and uses AI.

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